Record low runoff efficiency in the Upper Coloardo during water year 2026

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Key Points:

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Abstract

Plain Language Summary

1 Background

- Colorado river flow ("runoff") is very important for the western USA.
- Mass balance dictates, Runoff = Precipitation - Evaporation +/- change in storage.
- We normally can assume +/- change in storage is small.
- The year to year variation in precipitation drives a substantial amount of the variability in runoff.
- Emerging evidence suggests an *increasing sensitivity of runoff to air temperature* (e.g., Woodhouse et al. (2016)).
- Temperature is related to the Evaporation term in a number of direct and indirect ways, likely mediated through snow (Milly et al., 2018; Milly & Dunne, 2020).
- Annual runoff anomalies are commonly modelled as $\Delta Q = \alpha \Delta P + \beta \Delta T$ where $\Delta X = \frac{X - \bar{X}}{\bar{X}}$ for precipitation and runoff, and $\Delta T = T - \bar{T}$ (Lehner et al., 2017; Milly & Dunne, 2020). The "bar" symbol denotes the long term mean.
- Thus, we can estimate α and β based on observed data of precipitation, temperature, and runoff.
- For this work, I would also like to include peak springtime SWE as another predictor —— which may exercise an independent effect on runoff but is itself highly correlated with P and T.
- **2026 was incredibly anomalous** – it was the warmest year on record, runoff was the third lowest ever measured, springtime snow storage was the lowest on record (Marshall et al., 2026), and most critically — precipitation was low but not extrordinarily low.

2 Goals/questions

- How anomalous was 2026?
- Did 2026 "break' models of runoff sensitivity to temperature, precipitation, and SWE? Or do pre-2026 estimates of α and β still well describe runoff in 2026?
- SWE was very low, but precipitation was not as unusual. Does the data support the hypothesis that the SWE anomaly had a clear effect (independent of P and T) on the runoff ratio in 2026?
- Based on these insights, how much more likely is WY2026 runoff by the end of century? (e.g., it's a 1 in 20 year event, by 2080 it's 1 in 5).

3 Results (highly preliminary)

3.1 Historical context of 2026

- Fig 3.2 shows the historical trends, since 1906, of temperature, precipitation, runoff, and runoff ratio.
- 2026 was the warmest on record (z=4.3), had the lowest runoff ratio, the third lowest streamflow, and the lowest peak SWE ever observed.
- We can apply the estimate the coefficients for the equation $Q' = \alpha P' + \beta T'$ based on the data in Fig. 3.2
- Table 1 shows the coefficients α and β fit to data from 1936. These is basically identical to the coefficients in Lehner et al. (2019), Fig. 3.

- An α of 1.37 means that a 1% increase in precipitation yields a 1.37 percent increase in runoff.
- A β of -10.9 means that a 1° degree increase in annual air temperature reduces runoff by 10.9%.
- Including the effect of SWE reduces the temperature and precipitation sensitivity substantially — α goes from 1.29 to 1.05 and β is also reduced.
- Including SWE also improves the R^2.
- β has a large confidence range, which seems bad.

Table 1. Regression coefficients (± 95% confidence interval) for runoff and spring SWE anomalies, WY1936–2026 ($n = 90$). R^2 is uncentered.

Response	Predictors	Eq.	α_{tot}	β_{tot}	γ	R^2
ΔQ	$\Delta P, \Delta T$	1	1.29 ± 0.20	-10.0 ± 4.2	–	0.75
ΔSWE	$\Delta P, \Delta T$	2	0.53 ± 0.25	-12.9 ± 5.1	–	0.44
ΔQ	$\Delta P, \Delta T, \Delta SWE$	3	1.05 ± 0.19	-4.1 ± 4.0	0.45 ± 0.14	0.83

3.2 Looking forward

- I am simply taking the coefficients α and β and using those to estimate Q anomalies based on projections of T and P from WUS-D3.
- I am not using WUS-D3's own "runoff" variable — rather just precipitation, temperature, and springtime SWE from the models.

- Runoff and runoff ratio are projected to decline.
- Including SWE reduces the rate of runoff decline versus P and T only.
- This is likely because of non-linearities between annual warming abd annual precipitation and spring SWE.
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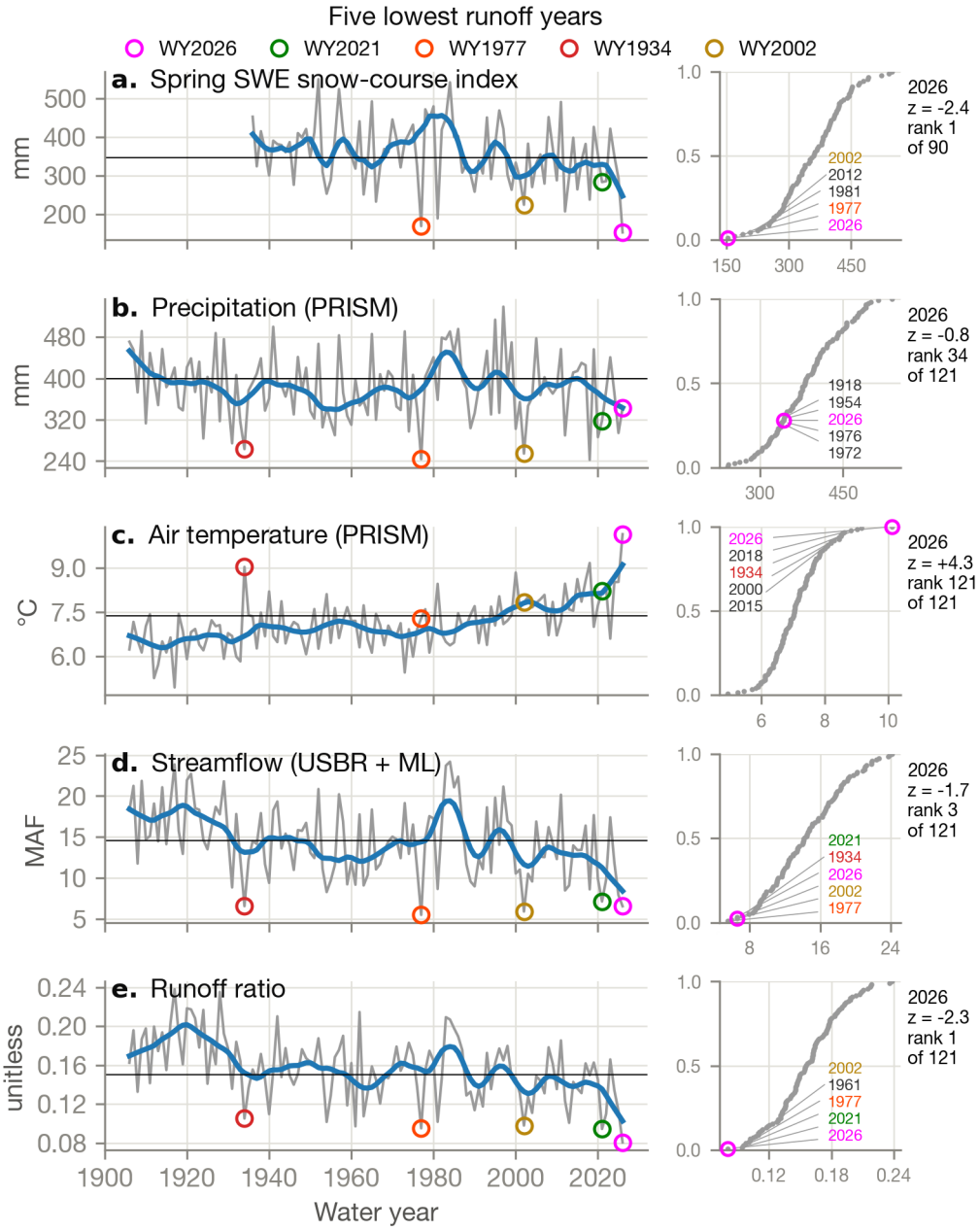


Figure 1. How anomalous was WY2026? Left: A timeseries of SWE, precipitation, temperature, streamflow, and runoff ratio (the ratio of total runoff to precipitation) for the Upper Colorado River at Lee's Ferry from 1904 to present. Snow surveys started in 1936. These data are all direct observations or based on observations. Right: the same, but the CDF of each variable. 2026 is highlighted in magenta and the four most similar years for each variable are displayed in text. The rank and z-score of 2026 is also displayed as text.

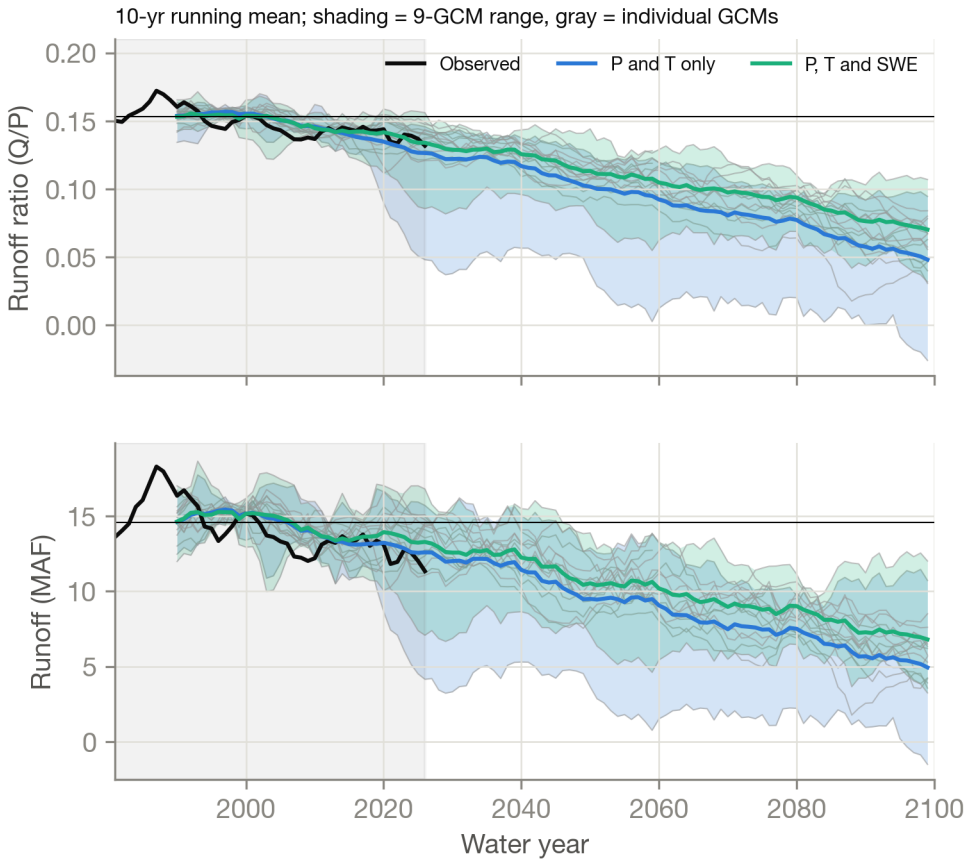


Figure 2. Projected changes in Runoff ratio (top) and Runoff (in million acre feet) under the SSP3-7.0 for 9 GCMs downscaled and bias-corrected by WUS-D3 (Table 2). Projections use WUS-D3 model P and T and the observationally constrained model of runoff sensitivity.

4 Appendix

5 Methods

We start from the decomposition framework used in several studies (Lehner et al., 2019; Milly & Dunne, 2020) (more), extended to include snow. We let

$$\Delta Q = \alpha \Delta P + \beta \Delta T + \epsilon_q, \tag{1}$$

where $\Delta X = (X - \bar{X})/\bar{X}$ for runoff, precipitation and SWE, $\Delta T = T - \bar{T}$, and the bar denotes the WY1981–2010 mean. All regressions are fit without an intercept.

Spring snow water storage (SWE) may exert an independent effect on runoff, but it is itself strongly correlated with precipitation and air temperature. We model the spring SWE anomaly similarly,

$$\Delta SWE = s_p \Delta P + s_t \Delta T + \epsilon_s, \tag{2}$$

where ϵ_s is the part of the SWE anomaly not explained by ΔP and ΔT. Substituting Eq. 2 into Eq. 1 gives

$$\Delta Q = (\alpha + \gamma s_p)\, \Delta P + (\beta + \gamma s_t)\, \Delta T + \gamma \epsilon_s + \epsilon_q. \tag{3}$$

In Eq. 1, α and β are the direct effects of precipitation and temperature with SWE held fixed. The coefficients in Eq. 3, $\alpha + \gamma s_p$ and $\beta + \gamma s_t$, are the total effects (α_{tot}, β_{tot}), including the part that acts through snow. These total effects should correspond to the precipitation elasticity and temperature sensitivity reported by Lehner et al. (2019) and Milly and Dunne (2020).

6 Previous literature

- Woodhouse et al. (2016) "In most recent drought years, low flows have been further exacerbated by warm temperatures". "Our work suggests warming temperatures are playing an increasingly important role in causing runoff declines even in nondrought years". They do not look at SWE, just precipitation, temperature, and soil moisture.
- Nowak et al. (2012). Looks at runoff efficiency explicitly. "Per degree warming, a 14% decrease in UCRB discharge was implied by the diagnosis of historical Lees Ferry streamflow and climate data." This might be the first study to show that temperature matters for runoff ratio in the UCRB
- Lehner et al. (2017) "Because the recent decades have also been marked by substantial upward temperature trends, the question of whether runoff ratio declines can be linked to temperature increases, and thus potentially to anthropogenic global warming, has gained the attention of the water management community". "... we estimate that in years with below-median precipitation, very low (¡10th percentile) runoff ratios are 2.5–3 times more likely if temperatures are warmer than normal (above-median)."
- Nolin et al. (2026)
- Palumbo et al. (2025)

7 Data and Methods

- **Runoff**: Lee's ferry naturalized streamflow (`https://www.usbr.gov/lc/region/g4000/NaturalFlow/provisional.html`).
- **Precipitation and temperature**: PRISM Long-term 1895-present (get citation). Monthly and annual precipitation total in the UCRB.
- **temperature**: NOAA USHCN monthly (station) data. This is a more homogenized dataset than PRISM. These are single stations only, but, we can compare

Table 2. CMIP6 GCMs dynamically downscaled in WUS-D3 (Rahimi et al., 2024) and used to drive the runoff ratio projections. FGOALS-g3 is excluded because of an artifact in its September precipitation.

GCM	Variant
ACCESS-CM2	r5i1p1f1
CanESM5	r1i1p2f1
CESM2	r11i1p1f1
CNRM-ESM2-1	r1i1p1f2
EC-Earth3-Veg	r1i1p1f1
EC-Earth3	r1i1p1f1
GISS-E2-1-G	r1i1p1f2
MIROC6	r1i1p1f1
MPI-ESM1-2-LR	r7i1p1f1

the anomaly to the PRISM air temperature anomaly. They are very similar. This bolsters the point that PRISM data is fit for purpose.

- **Peak SWE**: 1936 to present. I downloaded data from NRCS "COOP" snow course surveys. The data are downloaded from `https://wcc.sc.egov.usda.gov/awdbRestApi/swagger-ui/index.html`. There are 16 snow courses that have springtime survey data between 1936 to present.
- **Projections of temp, precipitation, and SWE**: WUS-D3. I'm using the bias-corrected version of (Rahimi et al., 2024) for 9 different GCMs under the SSP3-7.0 scenario with data from 1980 - end of century.

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